

Effect of XP-endo Shaper versus conventional rotary files on postoperative pain and bacterial reduction in oval canals with necrotic pulps: a randomized clinical study

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None group was associated with a significantly lower

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Introduction

Postoperative pain following root canal treatment continues to be a significant problem affecting the quality of lives of dental patients (Nixdorf *et al.* 2016, Evangelin *et al.* 2020). The initiation and progression patterns of postoperative pain are controlled by several variables that could be patient related such as anxiety, age, gender, tooth type, pulp vitality and periapical status; other factors involve microbial, chemical, and/or mechanical insults (Walton 2002, Azim *et al.* 2017a). Apical periodontitis, which is the primary cause of postoperative pain, could develop as a sequel to microbes left within root canal system, or iatrogenic extrusion of bacteria and their by-products, infected debris and/or irrigating solutions into the periapical tissue (Haapasalo *et al.* 2003, Siqueira 2003). Therefore, the main goal of root canal treatment is to efficiently eliminate root canal infection along with limiting the unfortunate occurrence of postoperative symptoms.

Oval root canals impose greater challenges for adequate preparation and disinfection. The prevalence of oval canals is high even apically and has been reported to exceed 50% (Wu *et al.* 2000). Studies using micro-computed tomography have shown that none of the available instrumentation systems could totally prepare nor eradicate bacteria from oval canals due to the discrepancy between the crosssectional shape of the canal and the round shape of instruments, leaving 32.38–79.9% of untouched canal walls with debris and bacterial biofilm (Paqué *et al.* 2010, Zuolo *et al.* 2018). There is a dilemma whether attempting to increase the apical size of canals by removing more contaminated dentine will predispose the root to dentine cracks and weakening, and the root canal to risks of stripping, zipping and transportation (Debelian & Trope 2015, Nathani *et al.* 2019, Pawar *et al.* 2019).

Recently, a different instrumentation concept has been described to prepare oval canals where the working width of the canal first described by Jou *et al.* (2004) has been considered. Specific instruments have been developed including the Self-Adjusting File (SAF, ReDentNOVA, Ra'anana, Israel), TRUShape (Dentsply Tulsa Dental Specialties, Tulsa, OK, USA), and recently the XP-endo Shaper (FKG Dentaire SA, La Chaux-de-Fonds, Switzerland) which has been reported to adapt to the natural cross section of the canal to prepare its full extensions without

unnecessary dentine removal (Metzger 2014, Azim *et al.* 2017b, Velozo & Albuquerque 2019).

The XP-endo Shaper system consists of a single heat-treated instrument. Its tip has six cutting edges which can start preparation at size 15 up to size 30. It has a small core with 0.01 taper and a serpentine shape which can expand on rotation to reach a final canal preparation with 0.04 taper. The small core grants the file flexibility, cyclic fatigue resistance, as well as effective debris elimination coronally (Azim *et al.* 2017b, FKG Dentaire 2018, Silva *et al.* 2018, Velozo *et al.* 2020). On the other hand, the use of conventional rotary files has been associated with apical extrusion of contaminated debris (Ahn *et al.* 2016, Topçuoğlu *et al.* 2016) and subsequently postoperative pain (Saha *et al.* 2018, Martins *et al.* 2019). Therefore, the research question of this study was whether root canal instrumentation using XP-endo Shaper in comparison with conventional rotary instrumentation using iRaCe files would result in a difference in postoperative pain, and intracanal bacterial count in mandibular premolars with single oval canals and necrotic pulps.

Material and methods

Study design, settings and sampling

The study was designed as a superiority, parallel, randomized, double blinded clinical trial. It was approved by the institutional review board/ethical committee in the Faculty of Dentistry - Cairo University. The protocol was registered at www.clinicaltrials.gov (ID: NCT0328533). Patients signed a printed informed consent form after explanation of the treatment procedure and associated possible discomforts. Recruitment and completion of the operative procedures for the study participants were done by the principal investigator (R.S.) at the outpatient clinic of the Endodontic Department, Faculty of Dentistry, Cairo University in Egypt, from July 2019 to October 2019.

A pilot study was conducted on 10 patients. Using a statistical power of 80% and alpha error of 5%, the effect size was calculated to be (0.811) at 6 h postoperatively. Sample size was calculated as 25 subjects/group. This number was to be increased to a sample size of 30 subjects/group to compensate for losses during follow-up (20% more than the calculated). Sample size calculation was achieved using G*Power: Statistical Power Analyses and Sample Size

Calculation software, version 3.1.9.3 (Heinrich Heine University, Dusseldorf, Germany).

Eligibility criteria and diagnosis

The inclusion criteria for the participants were healthy men and women of 20–60 years reporting no to mild preoperative pain and having mandibular premolars with single oval root canals and necrotic pulps, and with (smaller than 3 mm) or without periapical radiolucency. The exclusion criteria were pregnant women, if analgesics or antibiotics had been consumed by the patient during the past 12 h, teeth that did not have a normal occlusal contact on verification using an articulating paper, association with acute periapical abscess, greater than 5 mm periodontal pockets, greater than grade I mobility, alveolar bone loss exceeding 50%, and nonrestorable teeth.

The diagnosis was confirmed through history of chief complaint, negative response to both thermal test and electrical pulp tests (Denjoy DY310 Dental Pulp Tester; Denjoy, Henan, China) and radiographic examination using the digital imaging system of Digora Optime intraoral periapical phosphorus sensor plate and software version 2.5 (Digora Optime; Soredex, Helsinki, Finland).

Randomization and blinding

A random sequence of numbers (1–60) was generated by a co-investigator (Y.S.) using computer software (<http://www.random.org/>) and arranged in two equal columns according to the instrumentation system (A and B). Eight folded numbered papers were packed in opaque envelopes. After opening the access cavity to confirming the diagnosis by absence of bleeding, each patient selected an envelope and was allocated either to the XP-endo Shaper group (A) or the iRaCe group (B). The patients and outcome assessors were blinded to the assigned treatment group throughout the study.

Preoperative pain assessment

Each patient was given pain scale chart (modified visual analogue scale) to record the presence and severity of pain before treatment. The modified VAS consisted of 10 cm line and was illustrated visually, verbally and numerically to the patients. Four categorical scores expressed the pain level: 1, None (0); 2, Mild (1–3); 3, Moderate (4–6); 4, Severe (7–10).

Root canal treatment procedures and sampling

Root canal treatment was completed in a single visit. The tooth was anaesthetized using an inferior alveolar nerve block by local anaesthesia of 1.8 mL of 2% Mepivacaine HCl with 1:100 000 epinephrine (Mepe-caine-L; Alexandria Company for Pharmaceuticals and Chemical Industries, Alexandria, Egypt). All caries and/or coronal restorations were removed completely with sterile burs and a rubber dam was applied. The tooth, the clamp, and rubber dam were disinfected with 3% hydrogen peroxide followed by a sterile cotton swab with 5.25% NaOCl. The access cavity was then prepared using another sterile round carbide bur and Endo-Z bur (Dentsply Tulsa Dental Specialties) under sterile saline irrigation. The operative field and pulp chamber were disinfected as previously and 5% sodium thiosulfate was used to neutralize the effect of NaOCl then dried using sterile cotton wool. Sterility was checked by collecting control samples from the external surface of the crown as well as the access cavity using sterile paper points and culturing them aerobically and anaerobically.

For the collection of pre-instrumentation root canal bacterial samples (S1), 0.02 mL of the transport media brain heart infusion (BHI) broth (Brain heart infusion broth CM1135; Oxoid Microbiology Product Ltd, Basingstoke, UK) were placed into the canal with a sterile syringe and three sterile paper points were consecutively placed for 1 min each to soak up the fluid in the canal to a level approximately 1 mm short of the radiographic apex, and transferred aseptically to sterile tubes containing 1 mL BHI broth.

Stainless steel size 15 K-files were used to confirm the patency of the root canals. Working length was determined using an electronic apex locator (Root ZX; J. Morita USA, Irvine, CA, USA) then confirmed radiographically to be 0.5 mm shorter than radiographic apex. In the intervention group, the Xp-endo Shaper single file (FKG Dentaire SA) was used for instrumentation in an endodontic motor (X-Smart; Dentsply Tulsa Dental Specialties) at a speed of 800 rpm and a torque of 1 Ncm starting at size 30, .01 taper. The file was operated in 5 up and down strokes of 3–5 mm amplitudes until reaching working length then 10 brushing strokes of 5–7 mm for the full length of the canal and an extra 45 s of instrumentation was continued at the working length to enlarge the canal to size 40, .04 taper (De-Deus *et al.* 2019, Velozo *et al.* 2020). After each 5 strokes, the canal was

irrigated using 3 mL of 2.5% sodium hypochlorite and EDTA gel (MD-Chelcream; Metabiomed Co. Ltd, Chungcheongbuk-do, Korea) was used as a lubricant. In the control group, mechanical preparation was completed using rotary iRace files sizes 15, .06 taper, 25, .04 taper, 30, .04 taper, and 40, .04 taper (FKG Dentaire SA) at a speed of 600 rpm and a torque of 1.5 Ncm and the canal was irrigated after the use of each instrument. All canals received the same volume of irrigant (30 mL) using a syringe with 27 gauge open ended beveled needle placed at 3 mm from the working length.

The canals were then dried using sterile paper points and flushed with 5 mL of 5% sodium thiosulfate to inactivate the NaOCl. The final preparation was then checked by adapting master cones of size 40, .04 gutta-percha (Gutta Percha Points; Metabiomed Co. Ltd) at the working length and a radiograph was taken to verify proper length. The post-instrumentation root canal samples (S2) were taken from the canals as described above. The microbiological samples were transferred to the laboratory in the Microbiology department, Cairo University within 20 min. Canal filling was completed using a modified single cone technique with a resin-based root canal sealer (AdSeal; Metabiomed Co. Ltd). The access cavities were restored with core-build up composite resin (MultiCore Flow; Ivoclar Vivadent, Schaan, Liechtenstein) and occlusal contacts were adjusted.

Postoperative pain assessment

Patients were asked to record the incidence and severity of their postoperative pain on the modified visual analogue scale (VAS) at 6, 12, 24 h, 2, 3, 4, and 5 days. The patients were prescribed Ibuprofen (400 mg) (Novartis Pharma, Cairo, Egypt) to be consumed in case of persistent moderate to severe pain. Patients recorded the incidence and number of analgesic tablets taken.

Bacterial culturing

Intracanal bacterial counts were determined using a culture technique. Samples in the transport medium were dispersed with a vortex mixer for 60 s. Serial 10-folds dilutions (up to 10^{-3}) were prepared in tubes containing BHI broth. For nonselective obligate and facultative anaerobic bacterial culture, 50 microlitres of these diluted samples were transferred using micropipette and spread over BHI agar plates enriched with

10% sterile defibrinated sheep blood (Brain heart infusion agar CM1136; Oxoid Microbiology Product Ltd) and cultured under aseptic conditions by placing in an anaerobic sealed jar with GasPak (Anaerogen gas pack AN0035; Oxoid Microbiology Product Ltd) and anaerobic indicator (Anaerobic indicator BR0055B; Oxoid Microbiology Product Ltd) followed by incubation for 5 days at 37 °C (Kvist *et al.* 2004). The resultant growth was visually quantified by counting the number of colony forming units per millilitre (CFUs/mL) under the microscope (Martinho *et al.* 2014). Only plates with 30–300 colonies were counted. The number of CFUs/mL was then transformed into actual counts based on the previously determined dilution factors.

Assessment of flare-ups

If swelling occurred, the patient was seen to assess the severity by a blinded assessor (Y.S.) on a four point swelling rating scale expressed as none (n); small swelling intraorally adjacent to the treated tooth (s); moderate (m); or large severe swelling visible extraorally (l) (Tsisis *et al.* 2003). The blinded assessor also determined if systemic antibiotics (Augmentin 625 mg/8 h/5 days; GlaxoSmithKline, Cairo, Egypt) or drainage were needed. The incidence of flare-up was recorded as a binary outcome (Yes/No) when patients reported moderate to severe postoperative pain and/or moderate to severe swelling that began 12–48 h after treatment and lasted at least 48 hours (Pickenpaugh *et al.* 2001).



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